

# VLE and flash calculations

After midterm · Session 1 of 6 · 180 minutes

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30 September 2026

# The question for today

How does pressure change the phase split at fixed temperature?

Before class: 3-page reading · 15–20 minutes

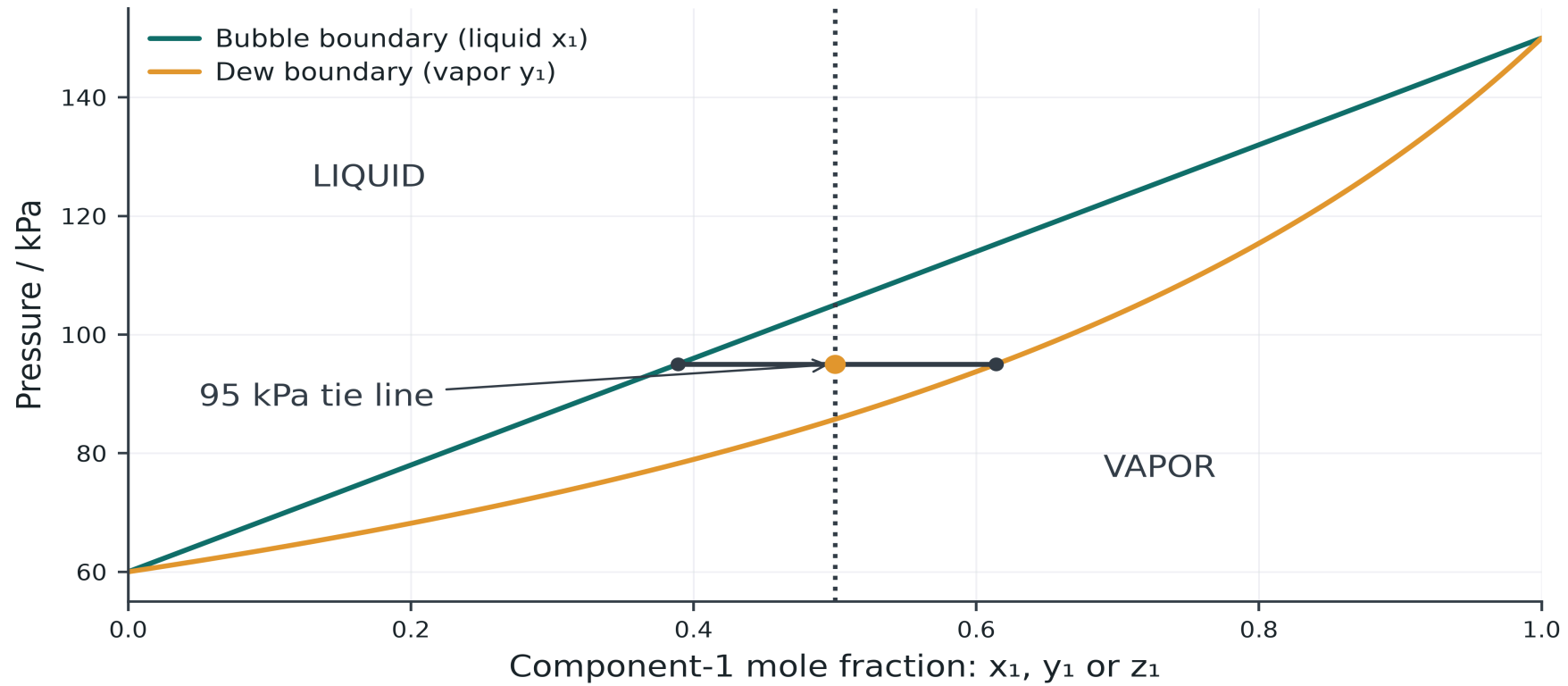
Bring the three preparation answers. Revisit your prediction after one controlled comparison and an independent check.

# The 180-minute class

| ACTIVITY                                | MINUTES |
|-----------------------------------------|---------|
| Opening question and essential concepts | 30      |
| One hand calculation                    | 20      |
| One lab demonstration                   | 20      |
| Break                                   | 10      |
| Student pair investigation              | 70      |
| Discussion, exit ticket and feedback    | 30      |

The 70-minute investigation is student work. Final block: discussion 20 min, exit ticket 5 min, feedback 3 min, closing 2 min.

# Read the phase diagram first



Synthetic ideal binary at 350 K. Horizontal: coexisting phases; vertical: fixed overall composition.

Predict the phase at 110 kPa before calculating.

# Boundaries and amounts answer different questions

$$P_{bubble} = \sum_i x_i P_i^{sat}, \quad \frac{1}{P_{dew}} = \sum_i \frac{y_i}{P_i^{sat}}$$

$$z_i = (1 - \beta)x_i + \beta y_i, \quad y_i = K_i x_i$$

The boundary equations assume ideal phases at fixed T. A TP flash supplies T, P and z, then finds compositions and amounts.

In a nonideal liquid, K depends on the unknown liquid composition.

# A reproducible ideal baseline

Lab 01 synthetic binary at 350 K:  $P_1^{sat} = 150$  kPa and  $P_2^{sat} = 60$  kPa. Choose ideal liquid,  $z_1 = 0.5$ ,  $P = 95$  kPa.

| QUANTITY               | EXPECTED RESULT    |
|------------------------|--------------------|
| Bubble P at $x_1=0.5$  | 105 kPa            |
| Dew P at $y_1=0.5$     | 85.7143 kPa        |
| Flash $x_1, y_1$       | 0.388889, 0.614035 |
| Vapor fraction $\beta$ | 0.493506           |

The values are synthetic definitions, not real-fluid measurements.

# **An independent check**

Independently close both component balances and confirm that all phase compositions sum to one.  
Explain the phase state using bubble and dew pressures.

# One demonstration • 20 minutes

Open the core lab

Use the synthetic ideal mixture at  $T=350$  K,  $P=95$  kPa and  $z_1=0.5$ . Save the baseline, then change only  $P$  to 110 kPa. Predict the phase before calculating.

Keep one saved baseline for the pair investigation.

# Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 01, save one ideal two-phase baseline and one changed pressure at the same  $T$  and  $z$ . Predict the change, report phase amounts and check one component balance by hand.

# Discussion • 20 minutes

How does pressure change the phase split at fixed temperature?

Compare results and discuss unexpected behavior. Use one numerical check to support your explanation.

# After class • 30–40 minutes

In Lab 01, save one ideal two-phase baseline and one changed pressure at the same T and z. Predict the change, report phase amounts and check one component balance by hand.

Assessment 1 and independent checkpoint

Retain one study JSON, one comparison and a short explanation.

Optional: Session appendix.

Learning path • Offline package

# Exit ticket + feedback · 8 minutes

## 5 min · Explain and check

Answer the session question in the form. Give one result or independent check from today's lab, with units.

## 3 min · Help improve the next class

Rate the pace, report lab access, and suggest one improvement.

Thai or English. No name, ID or email requested.

Wait for “**Response received.**” We will use the final 2 minutes to close the class.



Scan to submit Session 1 exit ticket and feedback.

[Open Session 1 form](#)